

SIMATS ENGINEERING

ASSESSMENT TOOL II — Visualization Analysis Report

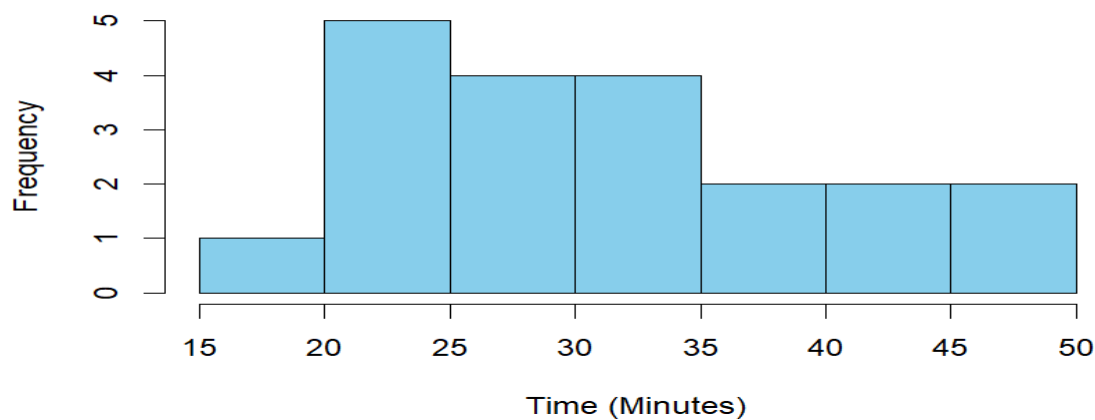
Code of Conduct:

I M.Arshini(192424322) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

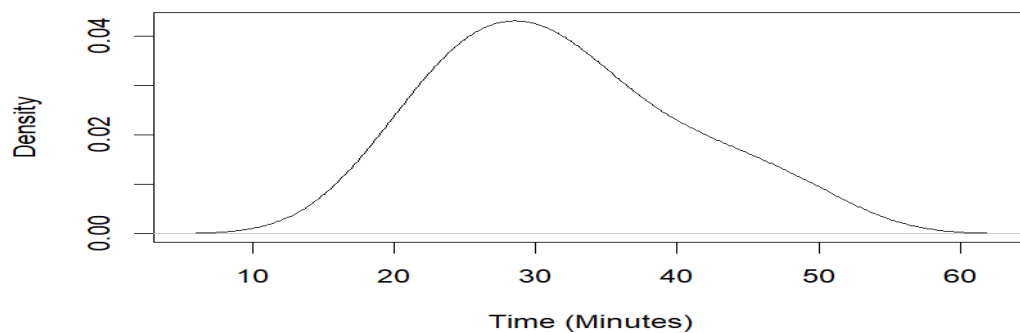
Signature of the student: G. Nithya Sree

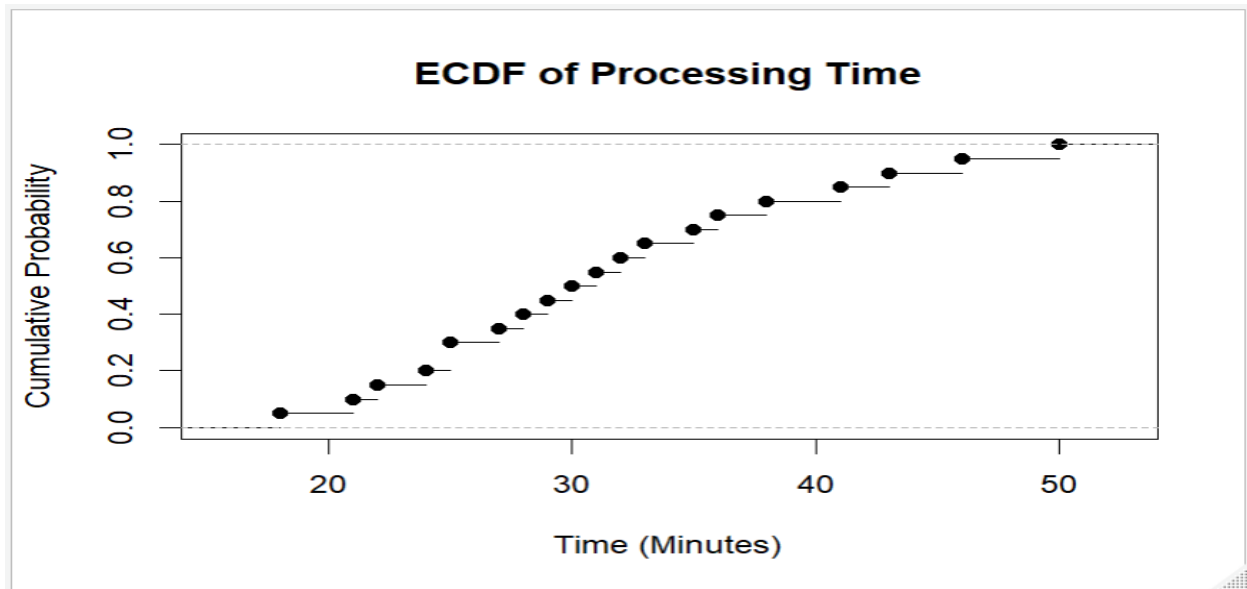
Q1. Comparing Distributions – Histogram, Density and ECDF

Histogram of Processing Time



Density Plot of Processing Time





CODE:

```
Time <- c(18,21,22,24,25,25,27,28,29,30,
          31,32,33,35,36,38,41,43,46,50)
hist(Time,
      main="Histogram of Processing Time",
      xlab="Time (Minutes)",
      ylab="Frequency",
      col="skyblue",
      border="black")
plot(density(Time),
     main="Density Plot of Processing Time",
     xlab="Time (Minutes)",
     ylab="Density")
plot(ecdf(Time),
     main="ECDF of Processing Time",
     xlab="Time (Minutes)",
     ylab="Cumulative Probability")
```

Short Interpretation

- **Histogram:** Shows the frequency of processing times. Most values are concentrated around **25–40 minutes**.
- **Density plot:** Shows the overall shape of the distribution more smoothly and helps identify **center and spread**.

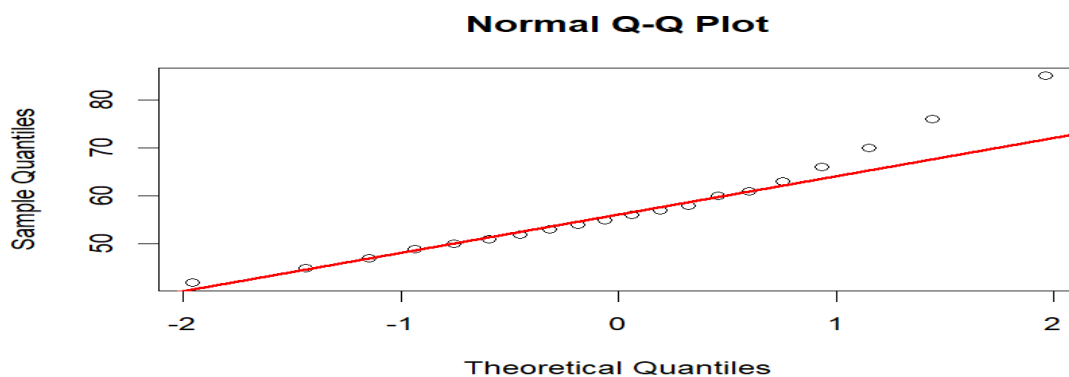
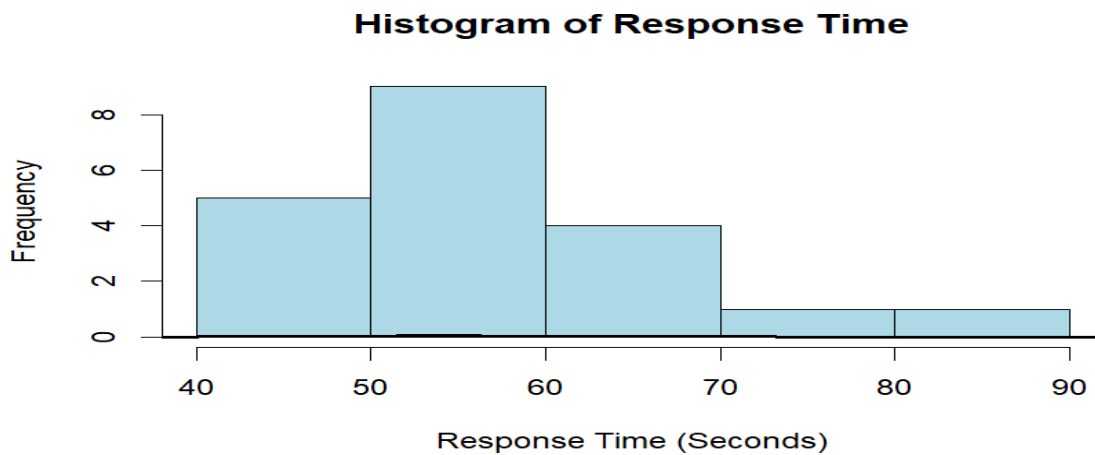
- **ECDF:** Shows the cumulative proportion of tasks completed at or below each processing time.

Comparison

Visualization	Best Shows
Histogram	Frequency, spread and approximate skewness
Density Plot	Overall shape, center and smooth distribution
ECDF	Cumulative behavior and percentiles

Conclusion: The data is centered around approximately 30–35 minutes, with a moderate spread. The higher values such as 46 and 50 create a slight right-side tail.

Q2. Q-Q Plot and Normality Analysis



CODE:

```
Response <- c(42,45,47,49,50,51,52,53,54,55,
             56,57,58,60,61,63,66,70,76,85)
hist(Response,
     main="Histogram of Response Time",
     xlab="Response Time (Seconds)",
     ylab="Frequency",
     col="lightblue",
     border="black")
lines(density(Response), lwd=2)
qqnorm(Response,
     main="Normal Q-Q Plot")
qqline(Response,
     col="red",
     lwd=2)
```

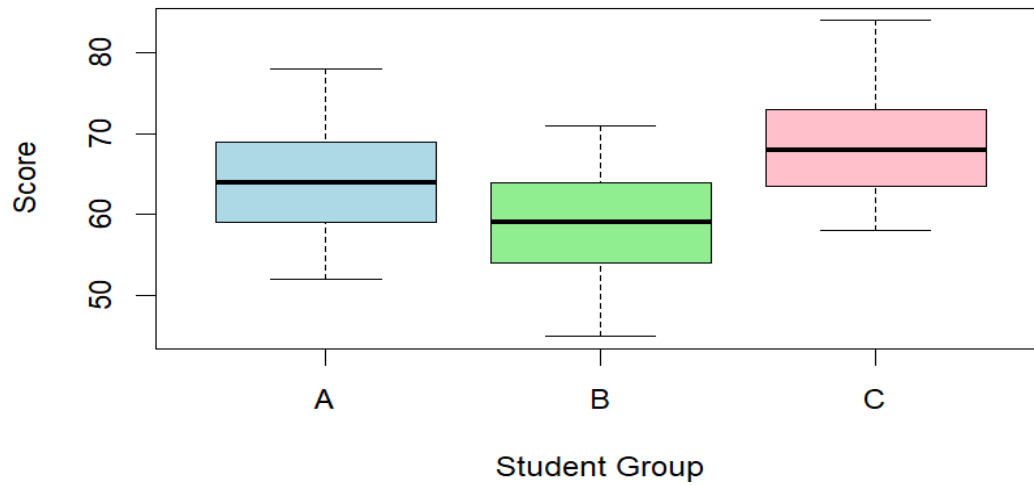
Short Interpretation

- If the points in the Q-Q plot follow the red line, the data is approximately normally distributed.
- Most middle observations are close to the reference line.
- The values 76 and 85 may deviate from the line at the upper end.
- This indicates a slightly heavier right tail or possible high-value observations.

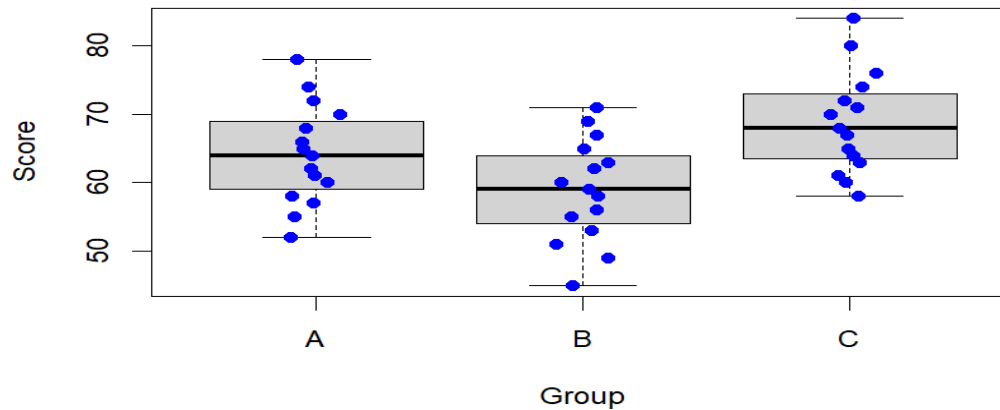
Conclusion: The response times are reasonably close to normal, but there is some departure from normality at the higher values.

Q3. Visualizing Many Distributions at Once

Comparison of Examination Scores



Comparison of Examination Scores



CODE:

```
GroupA <- c(52,55,57,58,60,61,62,64,65,66,68,70,72,74,78)
```

```
GroupB <- c(45,49,51,53,55,56,58,59,60,62,63,65,67,69,71)
```

```
GroupC <- c(58,60,61,63,64,65,67,68,70,71,72,74,76,80,84)
```

```
Scores <- data.frame(
```

```

Score = c(GroupA, GroupB, GroupC),

Group = rep(c("A","B","C"), each=15)

)

boxplot(Score ~ Group,

      data=Scores,

      main="Comparison of Examination Scores",

      xlab="Student Group",

      ylab="Score",

      col=c("lightblue","lightgreen","pink"))

```

Short Interpretation

- Group B has the lowest scores overall.
- Group C has the highest scores and median.
- Group A lies between Groups B and C.
- The boxplots allow us to compare median, spread and possible outliers in one graph.
- There are no major extreme outliers.

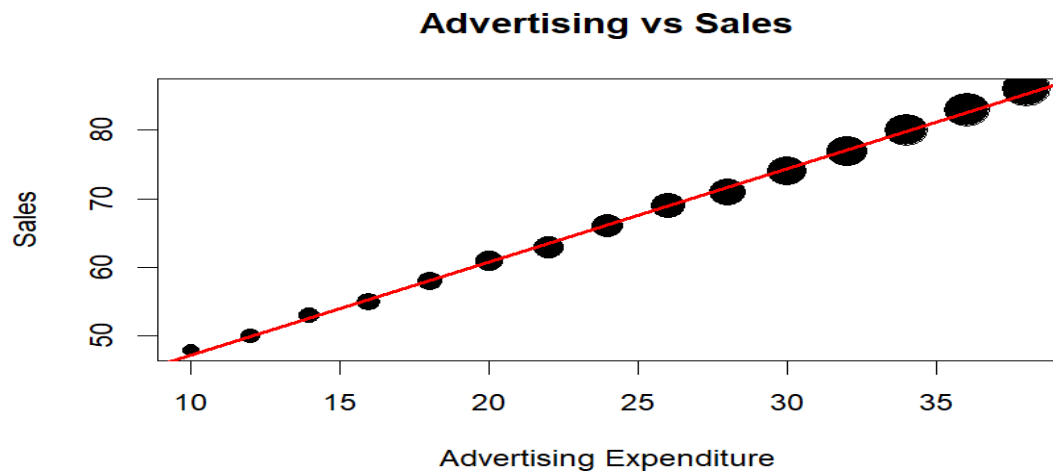
Why is one combined plot better?

A combined boxplot makes it easy to directly compare all three groups. Three separate graphs would make comparison of the medians and spreads more difficult.

Q4. Relationships and Multivariate Visualization

Here, we use:

- X-axis → Advertising
- Y-axis → Sales
- Point size → Website Visits



CODE:

```
Advertising <- c(10,12,14,16,18,20,22,24,26,28,
                30,32,34,36,38)
```

```
Visits <- c(120,135,150,165,180,195,210,225,
            245,260,275,290,305,320,340)
```

```
Sales <- c(48,50,53,55,58,61,63,66,
           69,71,74,77,80,83,86)
```

```
plot(Advertising, Sales,
     main="Advertising vs Sales",
     xlab="Advertising Expenditure",
     ylab="Sales",
     pch=19,
     cex=Visits/100)
```

```
model <- lm(Sales ~ Advertising)
```

```
abline(model,
```

```
col="red",  
lwd=2)
```

Short Interpretation

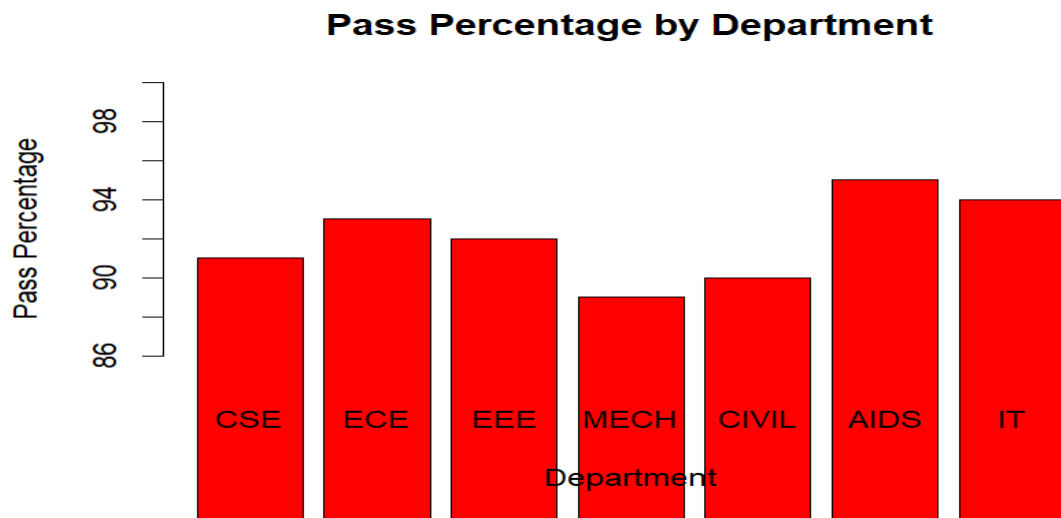
- The scatterplot shows a **strong positive relationship**.
- As advertising expenditure increases, sales also increase.
- The red trend line slopes upward, confirming the positive relationship.
- Larger points represent higher website visits.

Third variable analysis: Website visits provide additional information because higher advertising expenditure is also associated with more visits. However, too much visual encoding can make a graph cluttered. In this case, point size is simple and useful.

Q5. Misleading Visualization and Redesign

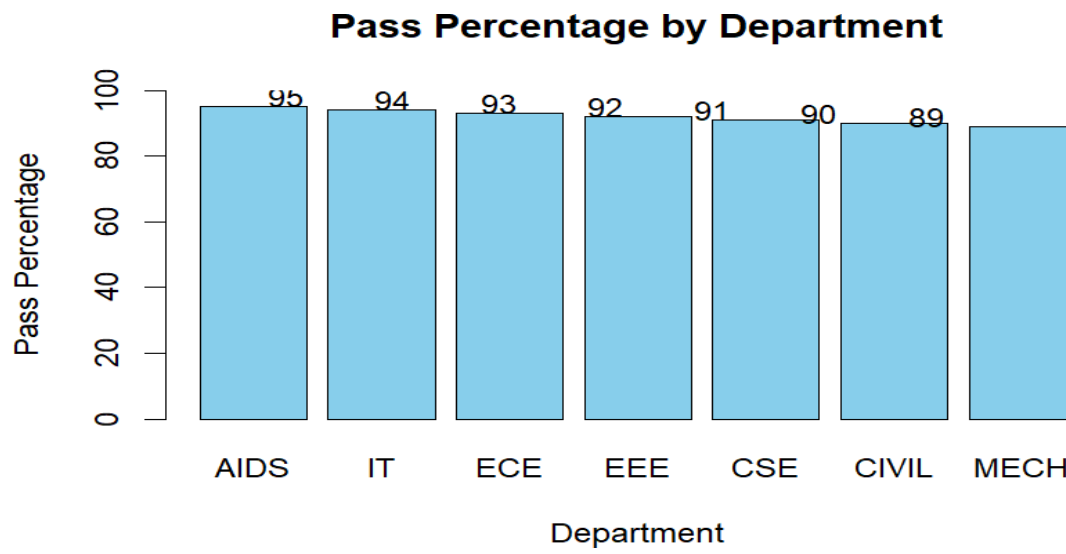
1. Misleading Visualization

This graph starts the Y-axis at 85 instead of 0, making small differences look very large.



Why is it misleading?

The actual values range only from 89% to 95%, a difference of just 6 percentage points. Starting the Y-axis at 85 exaggerates the visual difference between departments.



CODE:

```
Department <- c("CSE","ECE","EEE","MECH","CIVIL","AIDS","IT")
```

```
Pass_Percentage <- c(91,93,92,89,90,95,94)
```

```
barplot(Pass_Percentage,  
        names.arg=Department,  
        ylim=c(85,100),  
        main="Pass Percentage by Department",  
        xlab="Department",  
        ylab="Pass Percentage",  
        col="red")
```

```
order_data <- order(Pass_Percentage, decreasing=TRUE)
```

```
Department <- Department[order_data]
```

```
Pass_Percentage <- Pass_Percentage[order_data]
```

```
barplot(Pass_Percentage,
```

```

names.arg=Department,

ylim=c(0,100),

main="Pass Percentage by Department",

xlab="Department",

ylab="Pass Percentage",

col="skyblue")

text(x=seq_along(Pass_Percentage),

y=Pass_Percentage + 3,

labels=Pass_Percentage)

```

Interpretation

- **AIDS** has the highest pass percentage: **95%**.
- **MECH** has the lowest pass percentage: **89%**.
- All departments have relatively high pass percentages.

Comparison

Misleading Graph	Redesigned Graph
Y-axis starts at 85	Y-axis starts at 0
Small differences look large	Differences are shown honestly
Departments not ordered	Categories are ordered
Can mislead viewers	Easy and clear to understand

Conclusion: The redesigned graph communicates the data more honestly because it uses an appropriate baseline, ordered categories, and clear labels.